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Alcohol Use Severity among Hispanic Emerging Adults: Examining Intragroup Marginalization, Bicultural Self-Efficacy, and the Role of Gender within a Stress and Coping Framework

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ABSTRACT

Most research on cultural stressors and alcohol has focused on intercultural stressors. Continuing to exclude intracultural stressors (e.g., intragroup marginalization) from alcohol research will yield a biased understanding of the experiences of Hispanics living in a bicultural society. As we amass more studies on intracultural stressors, research will be needed to identify mutable sociocultural factors that may mitigate the association between intracultural stressors and alcohol. To address these limitations, we examined the association between intragroup marginalization and alcohol use severity and the extent to which gender and bicultural self-efficacy may moderate this association. A convenience sample of 200 Hispanic emerging adults ages 18-25 (men = 101, women = 99) from Arizona ($n = 99$) and Florida ($n = 101$) completed a cross-sectional survey. Data were analyzed using hierarchical multiple regression and moderation analyses. Higher intragroup marginalization was associated with higher alcohol use severity. Gender functioned as a moderator whereby intragroup marginalization was associated with higher alcohol use severity among men, but not women. Also, higher social groundedness functioned as a moderator that weakened the association between intragroup marginalization and alcohol use severity. Role repertoire did not function as a moderator. Our findings are significant because they enhance the reliability of the association between intragroup marginalization and alcohol use severity, and the moderating effect of gender in this respective association. This emerging line of research suggests that alcohol interventions targeting Hispanics may have a significant limitation by not accounting for intracultural stressors.

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Introduction

Epidemiological research in the United States (U.S.) has repeatedly found that alcohol is the most commonly used substance among *emerging adults*, a developmental stage that roughly spans the ages of 18 to 25 years.[1,2] Compared to adolescents and older adults, emerging adults are the age-group most likely to engage in alcohol risk behaviors (e.g., binge drinking, heavy drinking).[3] These trends are concerning because alcohol is involved in the top three leading causes of death (e.g., unintentional injury, homicide, suicide) among emerging adults.[4] Fields of developmental science propose that emerging adults are more likely to consume alcohol and engage in alcohol risk behaviors because this developmental stage is considered difficult and highly stressful due to transitional life changes, experiencing greater autonomy and high levels of instability, and taking on new and challenging developmental tasks.[5–6]

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Compared to other racial/ethnic groups, Hispanic emerging adults report the second-highest prevalence of current alcohol use (49.1%) and the highest prevalence of binge drinking (27.9%).[3] In part, this may be explained because in addition to normative developmental stressors, some Hispanic emerging adults disproportionately experience cultural stressors that may increase the probability of drinking alcohol and engaging in alcohol risk behavior.[7,8] *Cultural stressors* encompass interpersonal transactions and/or observations of the social environment that can be actual or perceived minor hassles or significant threats to well-being based on an individual's cultural characteristics that include, but are not limited to language, ethnicity, nationality, nativity/generation status, and migration experiences. The source of cultural stressors can occur between cultural groups (i.e., intercultural stressors) and within cultural groups (i.e., intracultural stressors).

To better understand alcohol use behavior among racial/ethnic minorities, more research is needed to identify and understand sociocultural determinants of alcohol use to inform the development of evidence-based interventions for racial/ethnic minorities.[9] From a public health perspective, more alcohol studies are needed with Hispanic emerging adults because this is an understudied population that reports high rates of alcohol risk behaviors.[3] Also, compared to other racial/ethnic groups, Hispanics are more likely to report alcohol-related disparities such as alcohol-related injuries, alcohol-related legal problems, and chronic liver disease.[10–13] Accordingly, this study focused on Hispanic emerging adults and emphasized sociocultural factors that may be associated with alcohol use. Specifically, this study aimed to examine the association between intragroup marginalization and alcohol use severity and the extent to which gender and bicultural self-efficacy may moderate this association.

To examine these aims, we build on the general framework of the *Transactional Model of Stress and Coping*.[14] In the present study, intragroup marginalization is conceptualized as a cultural stressor and transaction that may be perceived as stressful. *Intragroup marginalization* is the perceived interpersonal distancing exhibited by members of the heritage culture (e.g., Hispanic culture) toward an individual perceived to develop characteristics of the receiving culture (e.g., U.S. culture).[15] In line with this framework, it is expected that higher intragroup marginalization would be associated with higher alcohol use severity which is a maladaptive outcome. However, the association between the transaction (i.e., intragroup marginalization) and maladaptive outcome (i.e., alcohol use severity) can be weakened with adequate coping resources. In the present study, a coping resource is *bicultural self-efficacy*, the perceived confidence a person has to function effectively both within the receiving culture and the heritage culture.[16,17] Bicultural self-efficacy is conceptualized as a coping resource because it is an indicator of biculturalism that is considered a cultural asset that promotes resilience in response to cultural stressors.[18,19] Although the Transactional Model of Stress and Coping does not specify a role for gender, research has shown that men and women may respond differently to stressors.[20,21] For instance, men are more likely than women to drink alcohol in response to stress.[20,21]

Numerous studies on sociocultural determinants of alcohol among Hispanics have focused on cultural stressors because this hypothesized link aligns with the stress and coping paradigm.[14,22,23] Alcohol research among Hispanic emerging adults, and Hispanics of other age-groups, has indicated that higher levels of cultural stress are associated with higher alcohol use and alcohol risk behaviors; however, most studies have focused on intercultural stressors such as ethnic discrimination and consequently little is known about intracultural stressors such as intragroup marginalization and their associations with alcohol.[7,24–27] To our knowledge, only three prior studies have examined intracultural stressors and alcohol use among Hispanics and these studies focused on intragroup marginalization. Two of the studies used a total score of multiple subscales of the family intragroup marginalization inventory and did not find an association with alcohol use severity in samples of Hispanic college students and late adolescents.[25,27] The third study also used the family intragroup marginalization inventory, but only use the subscale, *family accusations of assimilation*, a perception or encounter in which family members of the heritage [Hispanic] culture accuse an individual of being highly acculturated because he or she developed characteristics of the receiving [U.S.] culture.[15,28] This study found that higher levels of accusations of assimilation were associated with higher levels of alcohol use severity among Hispanic emerging adult college students, and gender functioned as a moderator that strengthened this respective adverse association among men compared to women.[28]

Within the context of Hispanic cultures, family accusations of assimilation and other indicators of family intragroup marginalization may be salient factors associated with alcohol use behavior because family cohesion/connectedness and familial social support are highly valued by many Hispanics.[29,30] This is worth noting because lower levels of family cohesion/connectedness are associated with higher levels of alcohol use severity among Hispanics.[31,32] In

summary, more studies are needed on intracultural stressors and alcohol to empirically demonstrate if intracultural stressors are associated with alcohol use behavior and merit consideration in alcohol interventions targeting Hispanics.

Research on stressors, including cultural stressors, and their associations with alcohol suggest that the potential moderating effects of gender should be considered. Studies have repeatedly found that men are more likely to use alcohol to cope with stress, an association that has also been found among Hispanics.[30,31,33] Among Hispanics, cultural stressors may have a stronger association with alcohol use among men because women may be less inclined to use alcohol as a coping strategy because drinking alcohol is often not considered socially acceptable among women.[34,35] Further, in response to stress, Hispanic men may be socialized to be more independent and less dependent on family for social support to conform with masculine gender norms.[36,37] Thus, intragroup marginalization arising from within the family may further discourage Hispanic men from utilizing their family as a coping resource which may increase the probability of alcohol use and alcohol risk behaviors in response to cultural stressors and the cumulative effects of stress.

As the field of alcohol research among Hispanics continues to amass more studies on intracultural stressors, research is needed to identify sociocultural factors that may mitigate the adverse association between intracultural stressors and alcohol. Some sociocultural constructs that may merit investigating are indicators of biculturalism such as bicultural self-efficacy because they are considered "protective" factors in Hispanic health research.[38,39] Bicultural self-efficacy is comprised of multiple components, two of these components are social groundedness and role repertoire. *Social groundedness*, the confidence a person has establishing and maintaining social networks with both the receiving and heritage cultures is thought to enhance an individual's ability to cope with the pressure of living in a bicultural society.[16,17] *Role repertoire*, the confidence a person has using or learning culturally appropriate behaviors in encounters with the receiving culture and/or the heritage culture, is thought to facilitate positive interactions and reduce conflict, with both cultural groups.[16,17] Using the same sample from the present study, a prior study found that higher levels of social groundedness were associated with lower alcohol use severity, but the association between role repertoire and alcohol use severity was not statistically significant.[40] This study expands our prior research by examining *if* and *how* social groundedness and role repertoire moderate the association between intragroup marginalization and alcohol use severity.

Based on the review of the existing literature, the following hypotheses were proposed. *Hypothesis one*: higher levels of intragroup marginalization will be associated with higher alcohol use severity. *Hypothesis two*: gender will moderate the association between intragroup marginalization and alcohol use severity, whereby intragroup marginalization will have a stronger association among men than women. *Hypothesis three*: components of bicultural self-efficacy, social groundedness and role repertoire, will moderate the association between intragroup marginalization and alcohol use severity, whereby higher levels of social groundedness and role repertoire will respectively weaken the adverse association between intragroup marginalization and alcohol use severity. Lastly, we conducted post-hoc analyses to test three-way interactions between intragroup marginalization, respective components of bicultural self-efficacy, and gender. However, given the scarcity of literature, specific hypotheses regarding potential three-way interactions were not advanced.

Methods

Procedure and participants

The present study was approved by the Institutional Review Board of Florida International University. The present analyses used data from a cross-sectional study with a sample of 200 participants from the *Project on Health among Emerging Adult Latinos* (Project HEAL). Quota sampling was used to recruit a convenience sample of participants in Maricopa County, Arizona and Miami-Dade County, Florida using various recruitment strategies (e.g., in-person, posting flyers, targeted emails). Prospective participants interested in the study contacted the project coordinator to be screened and given access to the online survey if they met the eligibility criteria. Inclusion criteria for participants included being between ages 18 and 25, self-identifying as Hispanic or Latino/a, being able to read English, and currently living in Maricopa County, Arizona or Miami-Dade County, Florida. Exclusion criteria were currently being pregnant or breastfeeding. Participants provided informed consent to participate in the study by using an electronic informed consent form. All measures in the survey were in English. The survey took approximately 50 minutes to complete, and participants were compensated with a $30 electronic Amazon gift card. More details on the procedures for Project HEAL are published elsewhere.[40]

Measures

Alcohol use severity

Alcohol use severity was measured using the Alcohol Use Disorder Identification Test,[41]which consists of 10 self-report items with varied response choices on a Likert-type scale ranging from zero to four. Summed scores range from zero to 40 with higher scores indicating higher alcohol use severity. A sample item is "Has a relative or friend or a doctor or another health worker been concerned about your drinking or suggested you cut down?" Cronbach's reliability coefficient for this measure was α = 90.

Intragroup marginalization

Perceived intragroup marginalization was measured with the three-item family accusations of assimilation subscale from the Intragroup Marginalization Inventory—Family Scale.[15] A sample item is, "Family members tell me that I act White." Participants responded to items in this measure using a seven-point Likert-type scale (1 = *never/does not apply*, 7 = *extremely often*) and higher summed scores indicate higher perceptions of accusations of assimilation. Cronbach's reliability coefficient for this measure was α = .87.

Bicultural self-efficacy

Two components of bicultural self-efficacy, social groundedness and role repertoire, were measured with the corresponding subscales from the Bicultural Self-efficacy Scale.[16] The social groundedness subscale consists of seven self-report items and a sample item is, "I have an extensive network of mainstream Americans as well as an extensive network of people from the same heritage culture as myself." The role repertoire subscale consists of three self-report items and a sample item is, "I am confident that I can learn new aspects of both the mainstream American culture and my heritage culture." Both subscales use a nine-point Likert-type scale (1 = *strongly disagree*, 9 = *strongly agree*). Higher sum scores indicate higher levels of bicultural self-efficacy for each respective subscale. Cronbach's reliability coefficients for social groundedness and role repertoire in this study were (α = 0.92) and (α = 0.77), respectively.

Demographic questionnaire

To control for potential confounding effects, the following self-reported sociodemographic variables were assessed and included as dichotomous covariates: gender, (0 = men, 1 = women), study site (0 = Florida, 1 = Arizona), partner status (0 = single, 1 = has a partner), nativity (0 = immigrant, 1 = non-immigrant), Hispanic heritage group (0 = other Hispanic heritage, 1 = Mexican heritage), student status (0 = current college student, 1 = non-college student), employment status (0 = unemployed, 1 = employed). Age and financial strain (1 = has more money than needed, 2 = just enough money for needs, 3 = not enough money to meet needs) were also included as covariates. It should be noted that biological sex assigned at birth was not assessed.

Statistical analysis plan

Descriptive statistics, including means and standard deviations, were computed for continuous variables, and frequencies and proportions were generated for categorical variables. Correlations between continuous variables (i.e., age) and ordinal variables (i.e., financial strain, role repertoire, social groundedness, intragroup marginalization, alcohol use severity) were estimated with a Pearson correlation coefficient, correlations between dichotomous variables (i.e., gender, study site, partner status, nativity, Hispanic heritage, student status, employment status) and ordinal/continuous variables were estimated with a point-biserial correlation coefficient. Correlations between two dichotomous variables were estimated with a Phi coefficient. Multicollinearity was assessed using two diagnostic indicators, tolerance and the variance inflation factor (VIF). It is recommended that the tolerance value be higher than .10 and the VIF value be lower than 10.[42]At the item-level, missing data for multi-item measures were below 2% and the expectation-maximization algorithm was used to handle missing values prior to computing each respective scale score. The variables from the demographic questionnaire had no missing data.

A hierarchical multiple regression (HMR) model was used to estimate the main effects of the predictor variables on alcohol use severity. Predictor variables were entered into the HMR model in a specified order so that each block of predictors contributed explanatory variance to the outcome variable (i.e., alcohol use severity) after controlling for the variance explained by the previous block of variables.[42] Predictor variables were grouped and entered into the HMR model in the following order: (a) demographic variables were entered in the first block, (b) bicultural self-efficacy components, social groundedness and role repertoire, were entered in the second block, and (c) intragroup marginalization was entered in the third and final block to determine the extent to which it uniquely predicted

alcohol use severity above and beyond the other predictors.

Moderation analyses were conducted using PROCESS v3.2 for SPSS.[43] PROCESS tests moderation by (a) performing a multiple regression to replicate the variance explained by all the predictor variables included in the HMR model, (b) estimating interaction terms between the focal predictor (e.g., intragroup marginalization) and the moderating variable (e.g., gender), and (c) estimating conditional effects in relation to alcohol use severity. To estimate standardized regression coefficients in PROCESS, the variables were standardized. The moderation analyses controlled for all variables in the HMR model that were not included in the respective interaction terms. Three-way interactions were also tested in PROCESS using the method described above; however, three-way interactions control for the effects of two-way interactions among the variables used in the three-way interaction.

Results

Descriptive analyses and diagnostics

Frequencies, proportions, means, and standard deviations for all study variables are presented by gender in Table 1. Across study site, 49.5% ($n = 99$) of participants were from Arizona and 50.5% ($n = 101$) were from Florida. Regarding Hispanic heritage groups, the following groups were represented: Mexican ($n = 88$, 44.0%), Cuban ($n = 33$, 16.5%), Colombian ($n = 24$, 12.0%), other South American ($n = 21$, 10.5%), Central American ($n = 20$, 10.0%), and Puerto Rican ($n = 9$, 4.5%). It should be noted that among participants of Mexican heritage, 86 were from Arizona and two were from Florida. Bivariate correlations for all study variables are presented in Table 2. Assumptions of multicollinearity were met because all tolerance values were higher than .10 and all VIF values were lower than 10.

Hierarchical multiple regression

Table 3 presents the regression coefficients from the HMR model used to estimate the main effects on alcohol use severity. Results indicate that 20.3% of the variance in alcohol use severity was explained by all predictor variables entered in the HMR model. The first predictor block included demographic variables and explained 14.1% of the variance in alcohol use severity, $R^2 = .141$, $F(9, 190) = 3.47$, $p \leq .001$. The second block added components of bicultural self-efficacy, social groundedness and role repertoire, which explained 3.3% of the variance in alcohol use severity, $\Delta R2 = .033$ $F(2, 188) = 3.74$, $p = .03$. The third and final block added intragroup marginalization, which explained 2.9% of the variance in alcohol use severity, $\Delta R2 = .029$, $F(1, 187) = 6.93$, $p = .01$. Standardized regression coefficients from the final

Table 1. Descriptive statistics for study variables (n = 200).

Variable	Women 99 (49.5)	Men 101 (50.5)	
	n (%)	*n* (%)	χ^2
Study Site			0.50
Arizona	53 (53.5)	46 (46.5)	
Florida	49 (48.5)	52 (51.5)	
Partner Status			0.02
Single	72 (70.6)	70 (71.4)	
Has Partner	30 (29.4)	28 (28.6)	
Nativity			4.15*
Immigrant	24 (23.5)	36 (36.7)	
U.S.-Born	78 (76.5)	62 (63.3)	
Hispanic Heritage			0.10
Mexican	46 (45.1)	42 (42.9)	
Other Hispanic Heritage	56 (54.9)	56 (57.12)	
Student Status			0.01
Current College Student	71 (69.6)	68 (69.4)	
Non-College Student	31 (30.4)	30 (30.4)	
Employment Status			0.44
Employed	82 (80.4)	75 (76.5)	
Unemployed	20 (19.6)	23 (23.5)	
	M (*SD*)	*M* (*SD*)	*t*-value
Age	21.20 (1.92)	21.41 (2.26)	0.72
Financial Strain	2.35 (.57)	2.24 (0.61)	−1.89
Role Repertoire	20.97 (3.77)	21.57 (4.12)	1.06
Social Groundedness	43.66 (10.92)	47.01 (9.90)	2.27*
Intragroup Marginalization	6.09 (4.62)	6.69 (5.03)	0.89
Alcohol Use Severity	4.73 (4.80)	5.29 (7.02)	0.66

*$p \leq .05$.

Table 2. Bivariate correlations for study variables.

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Age	–												
2. Gender	-0.05	–											
3. Study Site	0.17*	0.05	–										
4. Partner Status	0.24**	0.01	0.07	–									
5. Nativity	0.08	0.14*	0.34**	-.01	–								
6. Hispanic Heritage	0.23**	0.02	0.86**	.06	0.32**	–							
7. Student Status	0.30**	-0.00	0.02	.01	-0.11	0.00	–						
8. Employment Status	0.27**	0.05	0.40**	.09	0.16*	0.37**	0.24**	–					
9. Financial Strain	-0.02	0.09	0.07	.03	0.02	0.06	-0.19**	0.02	–				
10. Role Repertoire	0.11	-0.06	-0.02	.03	0.14	0.02	-0.14*	-0.14*	-0.02	–			
11. Social Groundedness	0.05	-0.16*	-0.21**	-.15*	-0.09	-0.17*	0.16*	-0.06	-0.12	0.44**	–		
12. Intragroup Marginalization	-0.08	-0.06	0.28**	-.16*	0.13	0.17*	-0.08	0.13	0.20*	-0.06	-0.02	–	
13. Alcohol Use Severity	0.16*	-0.05	0.31*	-.01	0.12	0.23*	-0.03	0.21	0.10**	-0.07	-0.22**	0.27**	–

Note:.
*$p \leq .05$.
**$p \leq .01$. Correlations between continuous/ordinal variables were estimated with a Pearson coefficient, correlations between a dichotomous variable and an ordinal/continuous variable were estimated with a point-biserial correlation coefficient, correlations between dichotomous variables were estimated with a Phi coefficient.

Table 3. Hierarchical multiple regression model with alcohol use severity as the outcome.

	Model 1			Model 2			Model 3		
Variable	*b*	*SE*	β	*b*	*SE*	β	*b*	*SE*	β
Block 1									
Age	0.42	0.22	0.15*	0.46	0.22	0.16*	0.49	.21	0.17*
Gender	-0.81	0.82	-0.07	−1.13	0.82	-0.10	-0.90	.81	-0.08
Study Site	4.86	1.60	0.41**	4.32	1.59	0.36**	3.22	1.62	0.27*
Partner Status	-0.95	0.92	-0.07	−1.35	0.92	-0.10	-0.95	.92	-0.07
Nativity	0.08	0.96	0.01	0.08	0.95	0.01	-0.05	.94	-0.01
Hispanic Heritage	−2.33	1.59	-0.19	−2.30	1.57	-0.19	−1.72	1.56	-0.14
Student Status	−1.11	0.97	-0.09	-0.70	0.99	-0.05	-0.55	.97	-0.04
Employment Status	1.54	1.13	0.11	1.57	1.12	0.11	1.39	1.11	0.10
Financial Strain	0.75	0.70	0.08	0.64	0.69	0.06	0.28	.69	0.03
Block 2									
Role Repertoire				0.03	0.12	0.02	0.06	.12	0.04
Social Groundedness				-0.12	0.05	-0.20**	-0.12	.05	-0.22**
Block 3									
Intragroup Marginalization							0.24	0.09	0.19**

Note: b = unstandardized coefficient, *SE* = standard error, β = standardized coefficient.
*$p \leq .05$.
**$p \leq .01$.
***$p \leq .001$;.
R^2 = 14.1% for Block 1, ΔR^2 change = 3.3% for Block 2, ΔR^2 change = 2.9% for Block 3.

HMR model indicate that higher social groundedness was associated with lower alcohol use severity (β = −.22, p = .01). By contrast, older age (β = .17, p = .02), living in Arizona (β = .27, p = .05), and higher intragroup marginalization (β = .19, p = .01) were associated with higher alcohol use severity.

Moderation analyses

The moderation analysis, corresponding with hypothesis two, indicated that gender had a statistically significant interaction with intragroup marginalization in relation to alcohol use severity (β = −.27, p = .05). Conditional effects show that higher levels of intragroup marginalization were associated with higher alcohol use severity among men (β = .32, $p \leq .001$), but not women (β = .05, p = .65). This moderating effect is shown in Figure 1.

Moderation analyses, corresponding with hypothesis three, indicated that social groundedness had a statistically significant interaction with intragroup marginalization in relation to alcohol use severity (β = −.15, p = .02). This moderating effect is shown in Figure 1. The Conditional effects indicate that intragroup marginalization had the strongest adverse association with alcohol use severity for those reporting low levels of social groundedness (1 *SD* below the mean; β = .36, $p \leq .001$), and the association between intragroup marginalization and alcohol use severity was also statistically significant but weaker for those reporting average levels social groundedness (β = .20, p = .01). The conditional effect of intragroup

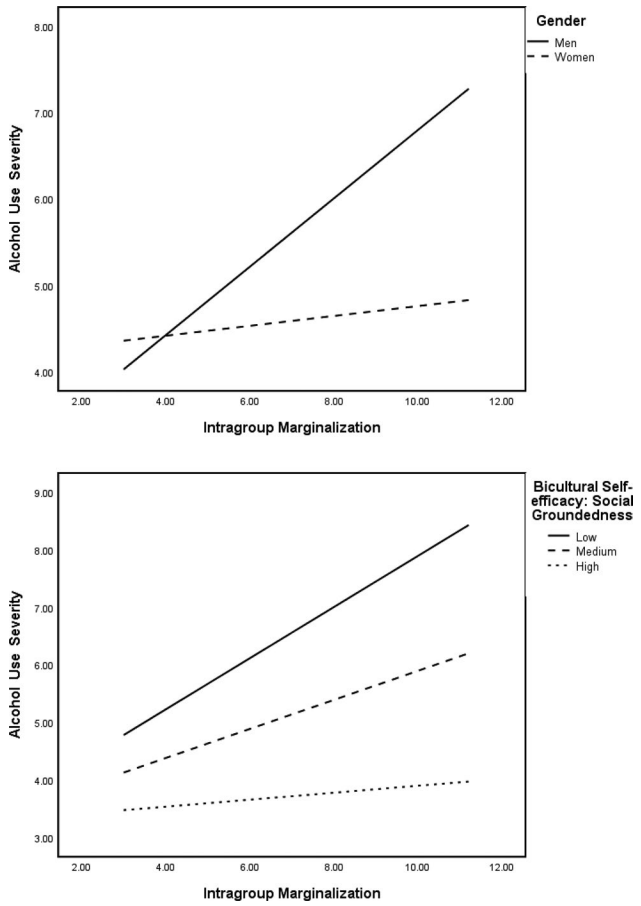

Figure 1. Two-way interactions with gender and social groundedness moderating the association between intragroup marginalization and alcohol use severity.

marginalization on alcohol use severity was not statistically significant at high levels of social groundedness (1 *SD* above the mean; β = .05, p = .09). Role repertoire did not moderate the association between intragroup marginalization and alcohol use severity (β = −.06, p = .32).

Results indicate that a three-way interaction between social groundedness, gender, and intragroup marginalization in relation to alcohol use severity was statistically significant (β = .32, p = .02). This moderating effect is shown in Figure 2. Conditional effects among men indicate that intragroup marginalization had the strongest adverse association with alcohol use severity at low levels (1 *SD* below the mean) of social groundedness (β = .55, $p \leq .001$), followed by the mean level of social groundedness (β = .31, p = .002). However, the conditional effect at high levels (1 *SD* above the mean) of social groundedness was not statistically significant (β = .08, p = .50). None, of the conditional effects among women were statistically significant.

Discussion

Prior research has established the adverse effect of intercultural stressors on alcohol use[23]—our study sheds new light by indicating that intragroup marginalization, an intercultural stressor, is robustly associated with alcohol use severity. Consistent with the Transactional Model of Stress and Coping, key findings from this study indicate that higher levels of intragroup marginalization were associated with higher levels of alcohol use severity. This association supports our first hypothesis and adds to the limited literature on intracultural stressors and alcohol use among Hispanics. Our second hypothesis was also supported and is consistent with prior research that the association between intragroup marginalization and alcohol use severity is stronger among men than women.[28] Our findings also partially support our third hypothesis and expand our understanding of the association between intragroup marginalization and alcohol use severity by identifying that social groundedness, a component of bicultural self-efficacy, mitigates this adverse association.

Our finding that intragroup marginalization is associated with alcohol use adds to the emerging evidence that intracultural stressors and their associations with alcohol use merit further investigation. Historically, a vast majority of research on cultural stressors and alcohol has focused on intercultural stressors; thus, continuing to exclude intracultural stressors from research will shed a skewed understanding of the experiences and stressors of Hispanics living in a bicultural society. One approach to change this trend is by expanding and/or developing new sociocultural theoretical/conceptual models of alcohol use that account for intracultural stressors.

Presently, only one theoretical/conceptual model, the *Bicultural Transaction Model* was found to propose a link between intracultural stressors and alcohol/substance use among Hispanics.[25] Furthermore, for the field of intracultural stressors and alcohol use to advance, studies with longitudinal designs will be needed to identify the mediating mechanisms to gain a deeper understanding of these associations.[44] For instance, social factors such as family cohesion/connectedness may function as a mediating mechanism due to its association with alcohol use severity and its plausible susceptibility to family intragroup marginalization.[28,32] Future research may also benefit from examining intrapersonal factors such as ethnic identity and self-esteem that may be affected by intragroup marginalization and have been associated with alcohol use behavior.[27] Identifying mechanisms that link

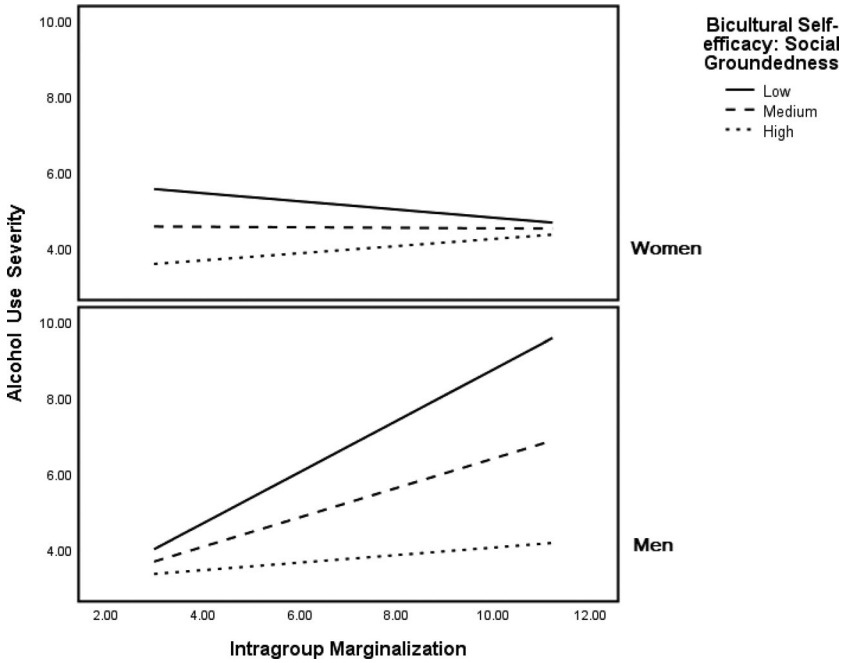

Figure 2. Three-way interaction with gender and social groundedness moderating the association between intragroup marginalization of assimilation and alcohol use severity.

intracultural stressors with alcohol will be essential to effectively inform and translate findings into evidence-based interventions.[44,45]

Our findings also highlight the need to examine gender as a moderator between intracultural stressors and alcohol use. Although men and women reported similar levels of intragroup marginalization our findings indicate that the association between intragroup marginalization and alcohol use severity was stronger among men than women—a finding that is consistent with a prior study.[28] Findings are also consistent with the *Theory of Gendered Prejudice* which posits that even at similar reported levels of cultural stress, men may be exposed to more egregious forms of intragroup marginalization/rejection which may have a stronger or longer-lasting impact.[46] To develop better explanations for these gender differences future studies could examine the interplay between intracultural stressors and gender roles (e.g., machismo, marianismo) in relation to alcohol use.[47,48] Also, integrating cognitive models of substance use may help identify distinct processes (e.g., drinking motives) between men and women that may mediate the association between intragroup marginalization and alcohol use behavior.[21]

Findings from our study suggest that bicultural self-efficacy, specifically social groundedness, is a cultural asset that may be functioning as a coping resource that buffers the adverse association between intragroup marginalization and alcohol use.[17,18] An explanation for this finding is that social groundedness may facilitate the ability to create and maintain broader and more diverse social networks with both cultural groups, which in turn, enhance social support.[49,50] From a stress and coping framework this is relevant because higher levels of social support can help Hispanics cope with cultural stressors more effectively; thus, mitigating their effects on substance use behavior such as alcohol use.[14,50,51]

The other component of bicultural self-efficacy, role repertoire, did not function as a moderator between intragroup marginalization and alcohol use severity. Considering the limited research on bicultural self-efficacy and alcohol, especially on specific components of bicultural self-efficacy, it is challenging to propose an explanation that is supported by existing literature.

Although the reliability of the role repertoire subscale was adequate in our study the psychometric properties of this subscale may have been a factor because the initial validation study reported suboptimal reliability and another study raised a concern about the psychometric properties of this subscale.[52] Some researchers have also speculated that some components of bicultural self-efficacy may be more important than others in relation to distinct outcomes.[17] This is a plausible explanation, particularly if bicultural self-efficacy is primarily associated with alcohol use via social support.

Regarding our findings from the three-way interactions, men may have benefited more from social

groundedness because (a) the association between intragroup marginalization and alcohol use severity was only statistically significant among men, (b) and compared to women, men reported significantly higher levels of social groundedness which may have helped to weaken the association between intragroup marginalization and alcohol use severity. By contrast, men and women reported similar levels of role repertoire; thus, it may be the case that men also need higher levels of role repertoire than women to offset the association between intragroup marginalization and alcohol use severity.

Findings from the present study should be interpreted considering the following limitations. First, we relied on self-report measures that are vulnerable to participant error and social desirability bias. Second, given the cross-sectional design of the study, we cannot make conclusive inferences regarding the causal or directional ordering of the associations observed. Third, limits to generalizability need to be acknowledged due to our non-probability sampling technique and the English-based survey utilized in the current study. Most participants were college students and U.S.-born, and all were English-speaking. Lastly, due to our sample size, we could not examine potential subgroup variations attributable to different Hispanic heritage groups.

Conclusions

Our findings are important because they replicate prior research; thus, enhancing the reliability of the association between intragroup marginalization and alcohol use severity, and the moderating effect of gender in this respective association. This study is also likely the first to identify a mutable moderating factor, social groundedness, that mitigates the adverse association between intragroup marginalization and alcohol use severity. Continuing to identify and examine understudied constructs associated with alcohol use among Hispanics will require a concerted effort to systematically advance our understanding of sociocultural determinants of alcohol use among Hispanics, and translating findings into efficacious and novel interventions.

Perhaps, equally important is our focus on emerging adults. Emerging adults are an understudied age-group in prevention science due to a focus on preventing and delaying the initiation of alcohol use among children and adolescents.[53,54] However, emerging adulthood merits extensive examination because developmentally this life stage affords opportunities to prevent and/or delay the onset of an alcohol use disorder and physical chronic diseases that may develop from prolonged heavy drinking by preventing the continuation and escalation of alcohol risk behaviors.[54–56] Thus, to design efficacious interventions that prevent the continuation and/or escalation of alcohol risk behaviors, more alcohol research is needed during emerging adulthood among Hispanics.[53,57]

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